

ALGEBRA PRACTICE SET

Instruction: Solve each problem and simplify your answer. Show your complete solution.

I. SIMPLIFYING

1. $3x - (2x - 5) + 4$
2. $2a - [3a - (a - 4b)] + 5b$
3. $5x - [3x - [2y - 4x - 3y] + (5x - 2y)] - 7x + 2y$
4. $ab^2 / (a^2b^{-1})$
5. $(2a^{-2}b^3 / ab^{-1})^2$
6. $(27a^{-3}b^6 / 8a^6b^{-3})^{-2/3} \cdot (a^4b^{-1} / a^{-2}b^2)$

II. EXPANDING

7. $3a^2b(2a - 4b + 5ab)$
8. $(2x - 5)(3x + 4)$
9. $(2x - 3y + z)(x + 4y - 2z)$
10. $(3x + 4)^2$
11. $(2x - \sqrt{y})(2x + \sqrt{y})$
12. $[(x + y) - 2][(x + y) + 2]^2$

III. FACTORING

13. $8x^3y - 12x^2y^2 + 4xy^3$
14. $ax + ay - bx - by$
15. $2x^3 - 3x^2y - 8x + 12y$
16. $x^2 - 3x - 10$
17. $6x^2 + 7x - 3$
18. $12x^2 - 5xy - 28y^2$

IV. RATIONAL EXPRESSIONS

19. $6x^2y / 9xy^2$
20. $(x^2 - 9) / (x^2 + 5x + 6)$
21. $(2x^3 - x^2 - 6x) / (x^3 - 3x^2 - 4x + 12)$

V. OPERATIONS ON RATIONAL EXPRESSIONS

22. $(x^2 - 4)(x + 3) - (x + 3)(x - 2)$

23. $[(x^2 - 1)(x + 2)] / [(x - 1)(x^2 - 4)]$

24. $[(x^2 - 9)/(2x^2 + 7x + 3)] \div [(x^2 - x - 12)/(2x^2 - 3x - 2)] - [(x^2 + 2x - 8)/(x^2 + x - 12)]$

VI. FRACTIONS AND COMPLEX RATIONAL EXPRESSIONS

25. $(3/4) \div (5/8)$

26. $(1 + 1/x) / (1 - 1/x)$

27. $(1/x - 1/y) / (1/x^2 - 1/y^2)$

ANSWER KEY

1. $x + 9$

2. b

3. $-14x + 3y$

4. b^3 / a

5. $4b^8 / a^6$

6. $4a^{12} / (9b^9)$

7. $6a^3b - 12a^2b^2 + 15a^3b^2$

8. $6x^2 - 7x - 20$

9. $2x^2 + 5xy - 3xz - 12y^2 + 10yz - 2z^2$

10. $9x^2 + 24x + 16$

11. $4x^2 - y$

12. $x^3 + 3x^2y + 2x^2 + 3xy^2 + 4xy - 4x + y^3 + 2y^2 - 4y - 8$

13. $4xy(2x - y)(x - y)$

14. $(a - b)(x + y)$

15. $(x - 2)(x + 2)(2x - 3y)$

16. $(x - 5)(x + 2)$

17. $(3x - 1)(2x + 3)$

18. $(3x + 4y)(4x - 7y)$

19. $2x / (3y)$

20. $(x - 3) / (x + 2), x \neq -3, -2$

21. $x(2x + 3) / [(x + 2)(x - 3)], x \neq -2, 2, 3$

22. $(x + 3)(x - 2)(x + 1)$

23. $(x + 1) / (x - 2), x \neq -2, 1, 2$

24. $-(x - 2)(5x - 21) / [(x - 4)(x - 3)(x + 3)]$

25. $6 / 5$

26. $(x + 1) / (x - 1), x \neq 0, 1$

27. $xy / (x + y), x, y \neq 0; x \neq \pm y$